

Bidirectional Applicability of Functional Logic Forward and Reverse Diagnosis of S, M, W Across Domains

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Abstract

This paper presents the bidirectional applicability of the Strüver Model's functional logic. The framework enables both forward prediction of system stability and reverse diagnostic reconstruction of its functional axes: Strength (S), Order (M), and Knowledge (W).

1 Introduction

The Strüver Model assumes that all stable systems are structured by three functional axes:

- Strength (S) – energetic or enforcement capacity
- Order (M) – regulatory coherence
- Knowledge (W) – informational structure

These axes generate stability $\Phi(t)$ when maintained within the dynamic calibration corridor.

2 Core Functional Equation

$$\Phi(t) = f(S(t), M(t), W(t))$$

Bidirectional applicability:

$$(S, M, W) \rightarrow \Phi(t)$$

$$\Phi(t) \rightarrow (S, M, W)$$

Reversibility follows from structural symmetry: stability cannot emerge without tri-
adic balance.

3 Reverse Diagnostic Method

Partial derivative sensitivity:

$$\frac{\partial \Phi}{\partial S'}, \quad \frac{\partial \Phi}{\partial M'}, \quad \frac{\partial \Phi}{\partial W}$$

Deviation magnitude indicates dysfunctional axis.

4 Domain-Specific Identification of S, M, W

The functional axes can be empirically mapped in each domain.

Domain	S (Strength)	M (Order)	W (Knowledge)
Physics / Energy	Energy flow, pressure gradients, temperature differentials	System regulation, structural integrity, constraint conservation	Feedback mechanisms, measurement coherence, control systems
Biology	Metabolism, muscular force, physiological vitality	Homeostasis, gene regulation, immune balance	Nervous processing, learning, sensory integration
Psychology	Willpower, motivational drive, action execution	Value hierarchy, impulse regulation, norm internalization	Conscious reflection, cognition, meta-awareness
Sociology	Power structures, resource allocation, enforcement capacity	Legal systems, moral frameworks, institutional stability	Communication networks, education systems, media coherence
Economics	Capital flow, production output, labor force capacity	Regulatory systems, compliance mechanisms, ethical constraints	Innovation density, research output, informational efficiency
Computer Science / AI	Computational power, activation density, processing throughput	Control logic, safety constraints, rule enforcement layers	Data structures, learning feedback loops, model coherence
Ecology / Climate	Primary productivity, energy transfer rates	Regulatory feedback loops, system damping	Biodiversity indices, adaptive information distribution

These mappings are empirically testable and allow structured functional diagnosis. Although domain manifestations differ, the structural logic remains invariant.

5 Functional Implications

The bidirectional structure enables:

- Predictive modeling of system behavior
- Diagnostic reconstruction of imbalance
- Cross-domain structural unification

The framework remains falsifiable, measurable, and domain-independent.

6 Conclusion

Functional logic is both predictive and diagnostic. Reverse application demonstrates that S, M, and W can be reconstructed whenever stability metrics are observable.

Thus, the model provides a universal structural diagnostic instrument across scientific domains.

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